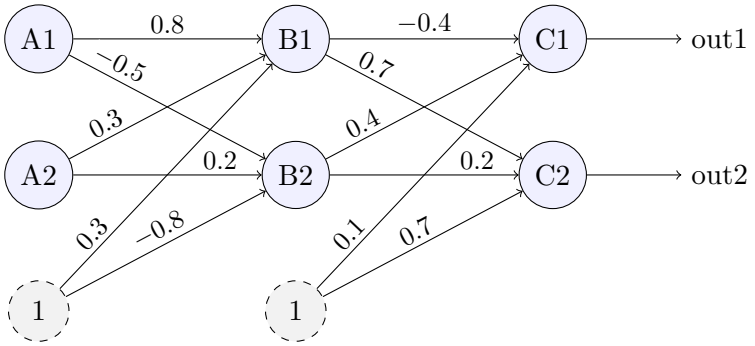
Forward propagation practice problem: solution



Problem. For the network above, out1 represents Class A and out2 represents Class B. Layer B uses the sign function as its activation, and layer C uses the softmax function. Show the forward pass, in matrix form, of the record with Feature1 = 5 and Feature2 = 10. How would this network classify the record (Class A or Class B)?

Notation. Row j of a weight matrix $\boldsymbol{W}^{(l)}$ holds the weights into node j of layer l: $W_{ji}^{(l)}$ is the weight from node i of the previous layer to node j. The bias of node j is $b_j^{(l)}$, the weight on the input that is always 1. Vectors are columns.

Solution

Weights, read from the diagram.

$$\boldsymbol{W}^{(1)} = \begin{pmatrix} 0.8 & 0.3 \\ -0.5 & 0.2 \end{pmatrix}, \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} 0.3 \\ -0.8 \end{pmatrix}, \qquad \boldsymbol{W}^{(2)} = \begin{pmatrix} -0.4 & 0.4 \\ 0.7 & 0.2 \end{pmatrix}, \quad \boldsymbol{b}^{(2)} = \begin{pmatrix} 0.1 \\ 0.7 \end{pmatrix}.$$

For example, the first row of $\boldsymbol{W}^{(1)}$ holds the weights into B1: 0.8 from A1 and 0.3 from A2.

A to B. The input is $\boldsymbol{x} = (5, 10)^\top$.

$$\boldsymbol{z}_B = \boldsymbol{W}^{(1)}\boldsymbol{x} + \boldsymbol{b}^{(1)} = \begin{pmatrix} 0.8 & 0.3 \\ -0.5 & 0.2 \end{pmatrix}\begin{pmatrix} 5 \\ 10 \end{pmatrix} + \begin{pmatrix} 0.3 \\ -0.8 \end{pmatrix} = \begin{pmatrix} 0.8 \cdot 5 + 0.3 \cdot 10 + 0.3 \\ -0.5 \cdot 5 + 0.2 \cdot 10 - 0.8 \end{pmatrix} = \begin{pmatrix} 7.3 \\ -1.3 \end{pmatrix}.$$

The sign function gives $+1$ for a positive input and -1 for a negative one:

$$\boldsymbol{B} = \operatorname{sign}(\boldsymbol{z}_B) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

B to C.

$$\boldsymbol{z}_C = \boldsymbol{W}^{(2)}\boldsymbol{B} + \boldsymbol{b}^{(2)} = \begin{pmatrix} -0.4 & 0.4 \\ 0.7 & 0.2 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 0.1 \\ 0.7 \end{pmatrix} = \begin{pmatrix} -0.4 \cdot 1 + 0.4 \cdot (-1) + 0.1 \\ 0.7 \cdot 1 + 0.2 \cdot (-1) + 0.7 \end{pmatrix} = \begin{pmatrix} -0.7 \\ 1.2 \end{pmatrix}.$$

The softmax turns these scores into probabilities. With $e^{-0.7} = 0.4966$ and $e^{1.2} = 3.3201$,

$$\boldsymbol{C} = \operatorname{softmax}(\boldsymbol{z}_C) = \frac{1}{e^{-0.7} + e^{1.2}}\begin{pmatrix} e^{-0.7} \\ e^{1.2} \end{pmatrix} = \frac{1}{3.8167}\begin{pmatrix} 0.4966 \\ 3.3201 \end{pmatrix} = \begin{pmatrix} 0.1301 \\ 0.8699 \end{pmatrix}.$$

Classification. out1 = 0.1301 (Class A) and out2 = 0.8699 (Class B). The larger output is out2, so the network classifies the record as **Class B**.